A fixed minimum-order threshold cannot explain the eleven omissions

2026-10-24 UTC. Post-observation, source-motivated audit.

The 17 November 1997 ANSI X9.62 draft permits an operator-supplied minimum subgroup order $r_{\min}$. For a prime full-order curve, the draft's test accepts exactly when $n \geq r_{\min}$. Across the four reconstructed fields with named omissions, any field-fixed minimum that admits all eight published controls also admits **seven of the eleven** omitted candidates. This excludes that one simple threshold as a complete selection rule, conditional on the original program having visited those positions.

The exact comparison

Let $m = \min(n_{\text{published}})$ within one field. To admit all published curves, $r_{\min} \leq m$. Every omitted curve with $n_{\text{omitted}} \geq m$ must then pass. A complete separator exists exactly when $\max(n_{\text{omitted}}) < m$.

Field	Published	Omitted	Unavoidable omitted ordinals
192-bit	3	2	5405
224-bit	1	3	777, 1485, 5736
239-bit	3	5	1562, 2491, 2682
256-bit	1	1	none

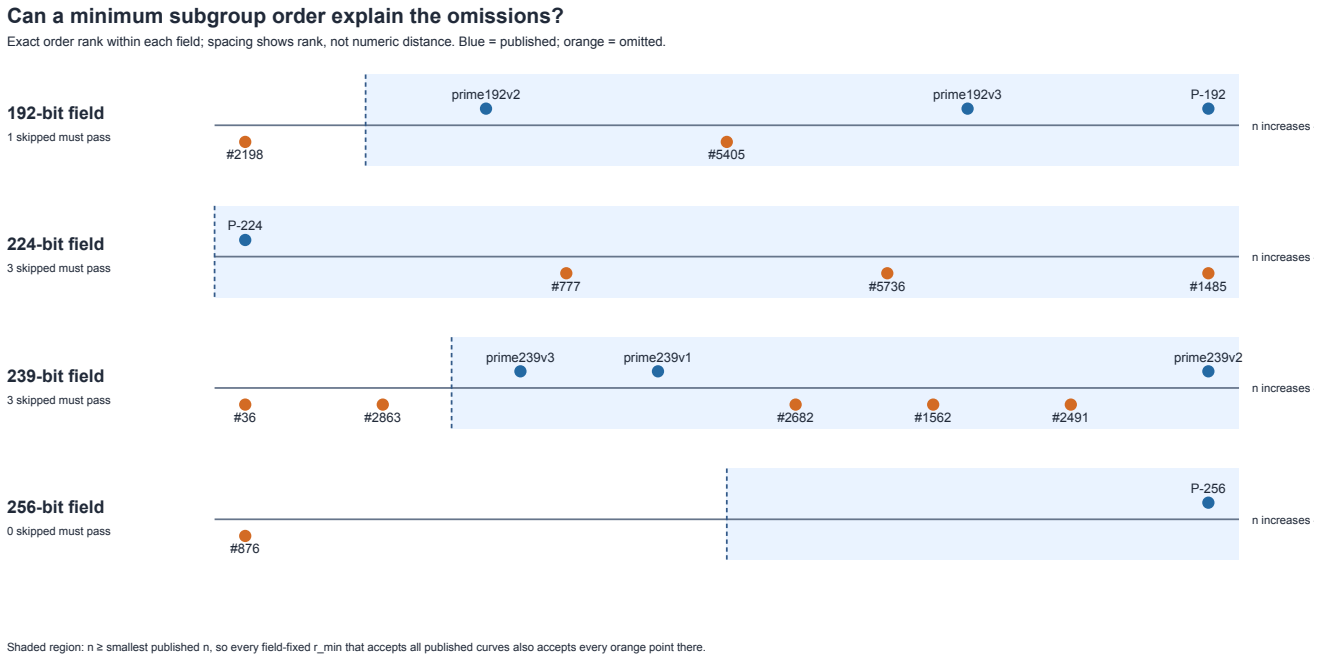


Figure 1. Within-field exact order ranks. Blue denotes publication; orange denotes an omission. Shaded positions pass every lower threshold that admits all published curves in that field. Spacing is rank rather than numeric distance. Source: analysis.json; editable source: render.py.

In the P-224 field, all three omitted orders exceed the published order. The 192- and 239-bit sets interleave. Only the P-256 pair permits a field-specific separating threshold; such a value would exceed the draft's advised upper bound $p + 1 - 2\sqrt{p}$, under which every full-prime-order candidate passes.

Evidence and limit

The source-pinned audit matched 19 exact (p, n) rows to prior prime-order point certificates. An independent script recomputed the pairwise inequalities from the archived omitted and published rows. This is not evidence that the 1997–98 program evaluated all eleven positions, used a fixed threshold, or rejected any for a weakness. A changing threshold or another test remains possible. The original program or candidate log is the decisive missing source.

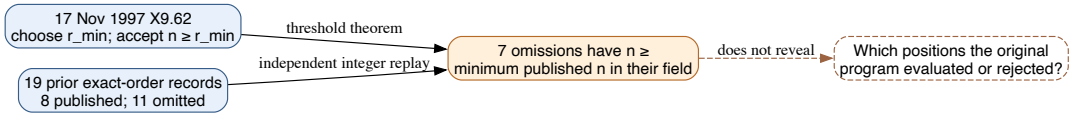


Figure 2. Solid edges are source or arithmetic checks; the dashed historical edge is unknown. Editable source: inference.dot.

Sources: ANSI X9.62 working draft, 17 November 1997, sections F.2.2 and F.3.2; hash-pinned curve, label and certificate records. Exact URLs, digests, method and limits: report.md, analysis.json and SCOPE.md.